# Regularized Density Ratio Estimation for Financial Time Series

B4 Research Progress Report — Late October 2026

Tanaka Hina

October 2026

Kanamori Laboratory – Department of Computer Science
Institute of Science Tokyo

# Background: Non-Stationarity in Financial Markets

- **Financial Context:** Asset return distributions fluctuate over time due to changing volatility regimes and macroeconomic shifts ($P_{\text{train}} \neq P_{\text{test}}$).
- **Covariate Shift Problem:** Traditional portfolio optimization assumes stationary distributions, leading to severe underestimation of tail risk during market stress.
- **Core Objective:** Estimate the density ratio $w(x) = \frac{p_{\text{target}}(x)}{p_{\text{source}}(x)}$ directly without estimating individual high-dimensional densities $p(x)$.
- **Quantitative Application:** Dynamic risk reweighting for Value-at-Risk (VaR) and Expected Shortfall (ES) estimations.

# Project Scope & Key Deliverables

- **Theoretical Focus:**
  - Analyze $L_2$-regularized density ratio estimation under non-convex constraint sets.
  - Establish finite-sample error bounds using empirical process theory.
- **Algorithmic Focus:**
  - Develop a fast optimization scheme using Accelerated Proximal Gradient methods.
- **Empirical Focus:**
  - Evaluate out-of-sample portfolio Sharpe ratios and risk metrics on simulated non-stationary Gaussian processes and real asset return data.

# Formal Problem Formulation

- Let $x \in \mathbb{R}^d$ be a $d$-dimensional vector of asset log-returns.
- We observe independent samples from source and target domains:

$$\mathcal{D}_{\text{source}} = \{x_i^s\}_{i=1}^{n_s} \sim p_s(x), \quad \mathcal{D}_{\text{target}} = \{x_j^t\}_{j=1}^{n_t} \sim p_t(x)$$

- Model the density ratio $w_\theta(x) \geq 0$ via a linear combination of kernel basis functions:

$$w_\theta(x) = \sum_{k=1}^{b} \theta_k \phi_k(x) = \boldsymbol{\theta}^\top \boldsymbol{\phi}(x)$$

- **Objective:** Minimize the squared loss function:

$$J(\boldsymbol{\theta}) = \frac{1}{2} \int \left( w_\theta(x) - \frac{p_t(x)}{p_s(x)} \right)^2 p_s(x) dx$$

# Unconstrained Least-Squares Importance Fitting (ULSIF)

- **Empirical Loss Function (Kanamori et al., 2009):**

$$\hat{J}(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top \hat{H}\boldsymbol{\theta} - \hat{h}^\top \boldsymbol{\theta} + \frac{\lambda}{2}\|\boldsymbol{\theta}\|_2^2$$

  where:

$$\hat{H} = \frac{1}{n_s}\sum_{i=1}^{n_s} \boldsymbol{\phi}(x_i^s)\boldsymbol{\phi}(x_i^s)^\top, \quad \hat{h} = \frac{1}{n_t}\sum_{j=1}^{n_t} \boldsymbol{\phi}(x_j^t)$$

- **Analytical Solution:** $\hat{\boldsymbol{\theta}} = (\hat{H} + \lambda I)^{-1}\hat{h}$
- **Limitations in High Dimensions ($d \gg n$):**
  - Matrix inversion scaling $\mathcal{O}(b^3)$ becomes computationally prohibitive.
  - Absence of explicit non-negativity constraints ($w_\theta(x) \geq 0$) leads to negative density ratio estimates in extreme market tails.

## Non-Negative $L_1/L_2$ Regularized Formulation

- To enforce non-negativity and sparsity in high-dimensional feature spaces, we solve:

$$\hat{\theta} = \arg\min_{\boldsymbol{\theta} \geq \mathbf{0}} \left\{ \frac{1}{2} \boldsymbol{\theta}^\top \hat{H} \boldsymbol{\theta} - \hat{h}^\top \boldsymbol{\theta} + \lambda_1 \|\boldsymbol{\theta}\|_1 + \frac{\lambda_2}{2} \|\boldsymbol{\theta}\|_2^2 \right\}$$

- **Sparsity Benefit:** Selects informative regime-change indicators while setting irrelevant kernel weights to zero.
- **Convexity:** The objective function remains strictly convex when $\hat{H} + \lambda_2 I \succ 0$, guaranteeing a unique global minimum.

# Convergence Rate and Finite-Sample Bounds

**Theorem 1 (Informal)**
Under sub-Gaussian assumptions on asset return distributions $p_s(x)$ and $p_t(x)$, for any $\delta > 0$, with probability at least $1 - \delta$:

$$\|w_{\hat{\theta}} - w^*\|^2_{L_2(p_s)} \leq \mathcal{O}\left(\frac{b \log(b/\delta)}{\min(n_s, n_t)}\right) + \mathcal{O}(\lambda_1 \|\boldsymbol{\theta}^*\|_1)$$

- **Key Insight:** The generalization error is dominated by the sample size of the target distribution $n_t$, which is typically small during rapid market regime shifts.

# Outline of the Proof Structure

- **Step 1 (Rademacher Complexity):** Bound the empirical process supremum $\sup_{\boldsymbol{\theta}\in\Theta} |\hat{J}(\boldsymbol{\theta}) - J(\boldsymbol{\theta})|$ using bounded kernel properties $\|\boldsymbol{\phi}(x)\|_\infty \leq M$.
- **Step 2 (Strong Convexity):** Leverage the lower bound on the minimum eigenvalue of the empirical Gram matrix $\lambda_{\min}(\hat{H}) \geq \gamma > 0$.
- **Step 3 (Current Hurdle):** Tightening the constant factor in the concentration inequality when returns exhibit heavy tails (sub-exponential / Student-$t$ noise instead of sub-Gaussian).

# Fast Non-Negative Proximal Gradient Descent

- **Iterative Update Rule:**

$$\boldsymbol{\theta}^{(k+1)} = \mathcal{P}_{\mathbb{R}_+^b}\left(\mathcal{S}_{\lambda_1\eta}\left(\boldsymbol{\theta}^{(k)} - \eta\left(\hat{H}\boldsymbol{\theta}^{(k)} - \hat{h} + \lambda_2\boldsymbol{\theta}^{(k)}\right)\right)\right)$$

- **Operators:**
  - $\mathcal{S}_\tau(u) = \text{sign}(u)\max(|u| - \tau, 0)$: Soft-thresholding operator ($L_1$).
  - $\mathcal{P}_{\mathbb{R}_+^b}(v) = \max(v, 0)$: Projection onto the non-negative orthant.
  - Step size condition: $\eta \leq \frac{1}{\lambda_{\max}(\hat{H}) + \lambda_2}$.

# Simulation Environment & Data Sets

- **Software Stack:** Python 3.11, NumPy, PyTorch (GPU acceleration for large Gram matrices $\hat{H}$), CVXPY (for baseline verification).
- **Synthetic Data Generation:**
  - Source $p_s \sim \mathcal{N}(\mathbf{0}, \Sigma_1)$ (Normal market conditions).
  - Target $p_t \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma_2)$ (Stressed market conditions with covariance shift).
- **Real-World Data:** S&P 500 ETF (SPY) daily and high-frequency return vectors (2020–2026).

# Empirical Evaluation on Synthetic Benchmarks

- **Convergence Speed:** The proximal gradient algorithm achieves $\epsilon$-accuracy in under 120 iterations across dimensions $d = 50, 100$.

| Method | Dimension ($d$) | MSE ($w_\theta$) | Computation Time (s) |
|---|---|---|---|
| Standard ULSIF | 50 | 0.142 | 0.08 |
| Constrained QP | 50 | 0.031 | 3.45 |
| **Proposed Prox-L1/L2** | 50 | **0.033** | **0.12** |

- **Observation:** The proposed method approaches Quadratic Programming (QP) accuracy while running $25\times$ faster.

# Importance-Weighted Value-at-Risk (VaR)

- **Risk Metric Reweighting:**

$$\mathrm{VaR}_\alpha = \inf \left\{ l \in \mathbb{R} : \frac{1}{n_s} \sum_{i=1}^{n_s} w_{\hat{\theta}}(x_i^s) \mathbb{I}(L(x_i^s) \geq l) \leq 1 - \alpha \right\}$$

- **Key Result:** Unweighted VaR underestimates 99% tail losses by 24% during regime transitions.
- Reweighted VaR using the estimated ratio $w_{\hat{\theta}}(x)$ captures market stress without waiting for new target samples to accumulate.

# Technical Challenges under Resolution

1. **Heavy-Tailed Distributions:** Financial returns exhibit high kurtosis. Standard sub-Gaussian concentration bounds are overly conservative.
    - *Proposed fix:* Replacing $L_2$ loss with robust M-estimation / Huber loss function.
2. **Bandwidth Selection for Kernels:** Hyperparameter tuning for $\sigma$ in Gaussian kernels $\phi_k(x) = \exp(-\|x - c_k\|^2/2\sigma^2)$ is sensitive to high-dimensional noise.

# Roadmap to Thesis Submission

- **November 2026:**
  - Complete robust loss extension for heavy-tailed noise.
  - Finalize Waterloo MFin / MMQF application documents.
- **December 2026:**
  - Conduct backtesting on full 10-year multi-asset portfolio dataset.
  - Complete draft of Bachelor's Thesis (*Sotsuron*) in English.
- **January–February 2027:**
  - Thesis submission (early February) and final oral defense (*Sotugyou Kenkyuu Happyou*).